

BRYAN KALEMBA

Computer Science Undergraduate | Aspiring Software Engineer & Web Developer

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PROFILE

Motivated problem-solver with hands-on experience building AI, computer vision, and web applications. Comfortable across Python, C, and Java, with a track record of delivering working software for real clients. Seeking a software engineering or web development internship / graduate scheme.

EDUCATION

BSc Computer Science, University of Nottingham

2025 – 2028 (Expected)

Relevant modules: Programming & Algorithms, Databases & Interfaces, Introduction to Software Engineering

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript

Tools & Concepts: APIs & API integration, Formspre, databases, algorithms, computer vision basics, voice-command interfaces, Git, version control, Linux

WORK EXPERIENCE

Website Manager (Voluntary)

Paula's Hair & Beauty Salon

- Managed and maintained the salon's website on a voluntary basis to gain professional web development experience, helping attract 20 new potential customers through the online booking system.
- Improved business efficiency by 25% by streamlining the appointment booking process through the website.
- Worked directly with the business owner to understand requirements and translate them into practical website features.

Customer Assistant

Vue Cinemas

- Helped improve sales and operational efficiency by 50% through proactive upselling and streamlined customer service.
- Delivered customer service in a fast-paced environment, developing communication, teamwork, and problem-solving skills.
- Resolved customer queries and complaints calmly and efficiently, helping maintain high customer satisfaction.

Volunteer

British Heart Foundation

- Contributed to improved sales performance by 50% through proactive customer engagement and support.
- Provided friendly, attentive support to customers, helping create a positive in-store experience.

PROJECTS

Jarvis AI Assistant

- Built a voice-controlled AI assistant for Windows using LLM function-calling (Gemini / local Ollama) to translate natural speech into ~25 real system actions — app control, file management, Steam integration, persistent memory.
- Designed a provider-agnostic architecture supporting both a cloud LLM and a fully local model behind one interface, with a schema converter reconciling their differing function-calling formats.
- Diagnosed and fixed a data-corruption bug from a non-JSON-serializable object failing to round-trip through storage, adding a self-healing recovery path.

Website for Paula's Hair & Beauty Salon

- Designed and built a business website using JavaScript, integrating Formspre to power an online booking form.
- Created an original design tailored to the salon's branding and themes, ensuring a cohesive look and feel across the site.
- Site was used by 20+ potential customers, directly supporting the salon's customer acquisition and booking process.

InkSync — Real-Time Collaborative Editor

- Built a Google Docs-style real-time collaborative text editor using Yjs, a CRDT library, so concurrent edits from multiple users merge automatically with no data loss.
- Hand-implemented the Yjs sync and awareness protocol over Socket.io instead of using the standard y-websocket library, to understand the actual handshake rather than treat it as a black box.
- Diagnosed and fixed a critical sync bug where new collaborators received a blank document, tracing the handshake message-by-message to the root cause.